Errors found, and how the design surfaced them

All 20 errors below are implemented as independent checks in `app/src/data/checks/run.ts` - plain code that recomputes an expected value from the raw packet (documents, extracted values, questionnaire, agency records) and compares it to the engine's `return_versions[v3]` output, the same way a reviewer would recompute by hand rather than trust the software. Every check appears in the **Checks** view with expected vs. actual, cited evidence, a plain-English explanation, and (where applicable) a proposed adjustment the preparer can accept or dismiss. Accepting one records an `adjustment` event and the return's workpaper total updates live.

Federal tax-effect rates (per dollar, from `return_versions[v3].tax_effect_rates`): stcg .348, ltcg .238, schc .355, interest .348, adjustment −.31, itemized −.26. California `ca_source_income` rate: .064.

Each check's dollar effect below is what accepting it alone moves, at today's baseline (no other checks yet accepted) - exactly the numbers the live app shows and `app/src/data/checks/run.test.ts` pins in a unit test. **Moves** says which workpaper total it hits: almost all of them are a **Tax** effect (AGI → total tax → balance due); the two payment checks (#15, #16) are a **Payments** effect instead - a dollar-for-dollar correction to estimated payments actually made, which never touches total tax, only the Payments line and therefore balance due (`balance due = total tax − payments`).

#	Check code	Error	Fed effect	CA effect	Moves	How the design surfaced it
1	`RSU-BASIS`	RSU sales taxed with $0 basis - the engine applied Corvane's supplemental-basis statement (D11) to the ESPP lot only, not the three RSU lots	−$15,536	−$2,765	Tax	The check reads D11's per-lot basis directly and compares it to the $0 basis baked into each F8949 lot's label. Two of the three lots (N1, N3) sold while still California-resident, so their California effect uses the `ca_source_income` rate (.064), not the blended `stcg` rate - the third (N4) sold after the move to Texas and uses the standard `ltcg` rate. Surfaced on **Home** (top of the ranked open-items list - this is the single largest effect in the file), **Checks**, and **Sales detail**'s exceptions-first section, which lists it before the 295-lot bulk table
2	`SCHC-1099K-DOUBLE`	Schedule C gross receipts double-counts a 1099-K ($12,400 Paywell amount is already part of the $41,200 Atlas deposit in the books)	−$4,402	−$384	Tax	The check sums 1099-NEC + non-1099 book revenue independently and compares to `SchC.1`. Surfaced on **Checks** and on **Books → tax**, which renders the revenue bridge with the duplicated line called out explicitly. This is also the item the 2028 IRS Letter 566 asks about - the **Filing** view's notice-response section answers it directly from this same check
3	`MILEAGE-DUPLICATE-PAGE`	A mileage log page is a second scan of the previous page, double-counting 3,700 miles	+$919	+$80	Tax	The check groups extracted trip rows by page, hashes each page's trips, and finds pages 2 and 3 identical - no hardcoded knowledge of which page is the duplicate. Surfaced on **Checks** and **Books → tax** (car-and-truck expense row)
4	`SE-HEALTH-INS`	Self-employed health insurance deducted despite Erik's eligibility for Ngozi's employer-subsidized family coverage	+$4,464	+$605	Tax	The check reads the questionnaire answer (qa-25) directly. Surfaced on **Checks**, with the 2024 return's matching treatment shown for context in the trace drawer on `Sch1.17`
5	`PERSONAL-USE-SPLIT`	Telephone/internet booked 100% business; a personal Spotify subscription booked as business software	+$541	+$47	Tax	The check applies call-01's own stated business-use percentages (phone 70%, internet 50%) and removes the named personal charge - disallowing an expense increases profit, so this is a positive effect. Surfaced on **Checks** and **Books → tax**

#	Check code	Error	Fed effect	CA effect	Moves	How the design surfaced it
6	`SEP-LIMIT`	The $18,000 SEP contribution exceeds the deductible limit, and the limit itself keeps moving as Schedule C is corrected	$0 at baseline	$0 at baseline	Tax (live)	This check is **live**: it recomputes the 20%-of-(profit – half SE tax) limit from today's workpaper Schedule C profit - at the unedited baseline the engine's $15,214 deduction already matches that limit exactly (so the effect of accepting it alone is $0), but the $18,000 actually contributed already exceeds it by about $2,786, and accepting errors #2, #3 and #5 first tightens the limit further and grows that excess in real time. Any excess is subject to a 6% excise tax on Form 5329 unless withdrawn. Surfaced on **Checks**, flagged on **Home** as a blocker
7	`HVAC-CAPITALIZE`	An $11,000 HVAC system replacement (with a permit) expensed as a repair instead of capitalized	+$3,633	+$896	Tax (what_if)	Too many downstream lines (depreciation, passive-loss release, AMT) for a single tax-effect-rate multiply, so this check surfaces the engine's own precomputed `what_if.hvac_improvement` alternative instead of inventing one. Surfaced on **Checks** and the dedicated **What-ifs** view, either of which can apply it
8	`NORTHLAKE-DUPLICATE-INTEREST`	$1,840 of sweep-account interest reported on both a composite 1099 and a standalone 1099-INT for the same account	–$640	–$127	Tax	Direct comparison of the two source amounts. Surfaced on **Checks** and **Reconciliations**
9	`SPARBANKEN-FX-YEAR`	Swedish interest converted at the 2024 average exchange rate instead of 2025's	+$43	+$9	Tax	The check re-divides the SEK amount by `fx_rates.SEK_per_USD["2025_average"]` and compares to the engine's figure, which cites the 2024 rate. Surfaced on **Checks**
10	`MERIDIAN-MISSING-LOT`	A 1099-B lot (page 32 of 32) was never extracted, leaving a $250 gap between the detail total and the summary	–$21	–$2	Tax	The check sums all extracted lot proceeds for the document and compares to the broker's own summary total extracted value - the gap itself is the evidence, no external knowledge of lot M18 required to detect it. Surfaced on **Checks**, **Sales detail** exceptions, and the **Reconciliations** box-total table
11	`CRYPTO-BASIS-MISSING`	A crypto lot has no cost basis or acquisition date, taxed as a $5,325 short-term gain	–$1,312	–$104	Tax	Flagged from the engine's own `DIGITAL_ASSET_BASIS_MISSING` diagnostic; the client's message (msg-07) supplies the missing basis and date. Surfaced on **Checks** and **Sales detail** exceptions; the demo worked file shows the question-to-client → answer → adjustment flow end to end
12	`MORTGAGE-PAYOFF-DOUBLE`	Payoff interest counted twice - the payoff statement says plainly it's already in the Form 1098 box 1 figure	+$559	+$103	Tax	Direct comparison of the two source documents' interest amounts. Surfaced on **Checks**
13	`PROPERTY-TAX-SELLER-CREDIT`	Home property tax includes the seller's pre-closing share (credited at closing), and the Oregon composite tax paid on Erik's behalf is omitted from the SALT total	+$1,051	+$332	Tax	The federal effect isn't the raw dollar correction - the check applies the $40,000 SALT cap first: the corrected uncapped total ($35,958) falls under the cap, so the federal deduction moves by the full $4,042 gap to the cap, not the $5,332 raw correction. California has no SALT cap but also never deducts state income tax at all, so only the real-estate-tax correction ($6,912) affects the California side - the Oregon composite tax is federal-only. Surfaced on **Checks**

#	Check code	Error	Fed effect	CA effect	Moves	How the design surfaced it
14	`VEHICLE-DONATION-LIMIT`	A donated vehicle claimed at the client's $6,000 KBB estimate instead of the $2,900 the charity actually received at sale	+$806	+$149	Tax	Direct comparison to Form 1098-C box 4c. Surfaced on **Checks**
15	`FED-ESTIMATED-PAYMENTS`	A federal Q4 estimated payment taken from the client's hedged recollection ("4,000 I think") instead of the IRS transcript	+$1,000	—	**Payments**	The check sums the IRS account transcript's own `670` transactions and compares to the claimed total - the $1,000 overclaim reduces the federal Payments line (total tax is untouched), raising balance due by the same amount. Surfaced on **Checks**, **Reconciliations**, **Filing** (transcript table), and **Home**'s top strip (the Payments row, not Total tax)
16	`CA-ESTIMATED-PAYMENT-MISDESIGNATED`	A California estimated payment was designated to the wrong tax year at the FTB and shouldn't count toward 2025	—	+$3,500	**Payments**	Compares the client's Web Pay confirmation to the FTB account record, which shows the payment held as a 2024 credit - it reduces the California Payments line, not Total tax. Surfaced on **Checks** and **Home**'s top strip
17	`CA-SOURCE-WAGES-PAYROLL-ERROR`	California-source wages use the full-year W-2 Box 16 instead of the actual CA work period (payroll kept coding CA after the move)	$0	−$5,685	Tax (`ca_source_income`)	Box 1 federal wages don't change at all - only how much of them is CA-source - so this is a pure `ca_source_income` reallocation with a zero federal effect by construction, not a reduced federal `wages` rate. The check compares Box 16 to the paystub's own year-to-date CA wages as of the move date, corroborated by the client's note (qa-45). Surfaced on **Checks**, **Home** blocker, and the **State split** view's flagged section
18	`THEO-SSN-TYPO`	A transposed SSN digit in the organizer caused a federal e-file rejection	no tax effect	no tax effect	—	Compares the organizer's entry to the Social Security card and the 2024 return. Surfaced on **Checks**; the rejection itself is visible in **Filing**'s e-file submissions table once the as-of clock passes 7/13
19	`LITTLE-SPROUTS-EIN-MISREAD`	A daycare provider EIN misread at 97% confidence (digits transposed)	no tax effect	no tax effect	—	Low-confidence extraction flagged directly. Surfaced on **Checks** and the document's own confidence badge in **Documents**
20	`BRIGHT-OAK-NAME-MISMATCH`	A care provider's name on Form 2441 doesn't match its own statement or the 2024 return	no tax effect	no tax effect	—	Direct string comparison. Surfaced on **Checks**

The combined effect, if every check is accepted

Home's top strip shows this live, as three columns per jurisdiction (engine v3 / workpaper as accepted so far / workpaper if every still-pending check were also accepted). Accepting all 20 checks moves:

	Federal	California
Total tax: v3 → all accepted	$81,962 → $72,068	$18,874 → $12,027
Payments: v3 → all accepted	$77,742 → $76,742	$27,130 → $23,630
Balance due: v3 → all accepted	$4,220 due → **$4,674 refund**	$8,256 refund → **$11,603 refund**

Total tax moves by the sum of the 18 tax-effect checks only; Payments moves by the two payment corrections only (#15, #16); balance due is always `total tax − payments`, recomputed, never shifted directly - which is exactly the bug a payment correction used to cause before `app/src/data/workpaperTotals.test.ts` pinned both totals down.

Judgment and completeness items (no engine fix, but recorded)

These don't have a single "correct" numeric answer - they're judgment calls, conflicts, or outside-filing items the brief requires proving were considered. They live in `data/judgmentItems.ts` and appear on the **Review notes & sign-offs** view, where a preparer records each one as a `judgment` event citing its authority (per the firm standard "every judgment call records the authority relied on"):

- **IRA basis conflict** - client says no prior basis; the 2024 Form 8606 and 5498 both show $7,000. Documents win; the engine's $26 taxable conversion is correct as computed.
- **St. Brigid receipt** - an ambiguous digit (41% confidence) resolved by a client message and a specific check number; a written acknowledgment is still required for gifts of $250+.
- **S corp basis** - rolled forward through a late K-1 to a $21,215 excess distribution (taxed as long-term gain), which the prior preparer's file simply "assumed... sufficient."
- **Prior-preparer open items** - two resolved by newly-received documents, one immaterial (home office square footage), one still open (Form 3520 for a 2023 inheritance, outside this engagement - advised in writing per Circular 230 §10.21).
- **Outside filings identified**: FBAR, Form 3520, two late 1099-NECs, Oregon/Arizona composite returns already filed by the entities, and Ada's own dependent return.
- **California AMT on the ISO exercise** - not computed by the engine; recorded as a judgment call given the exercise date fell after the move to Texas.
- **Questionnaire conflicts** - "no sales" (qa-14) vs. 282 actual sales; a wrong Q4 estimate (qa-41, see check #15).
- **Due diligence** - Form 8867 required for CTC/ODC/AOTC, tracked as a standing item.
- **Extension shortfall** - the April extension payment was sized from v2's estimate; the final balance differs, so interest and a failure-to-pay penalty apply to the late portion.

What the engine got right (passed checks)

Eleven checks pass today, shown alongside the failures in the same **Checks** view so a reviewer sees the full independent-verification picture, not just the bad news: W-2 de-duplication, the corrected Meridian 1099 superseding the original, 260 Brightpath lots reconciling exactly, excess Social Security, Form 8959, Form 8938's threshold, Form 8606's pro-rata math, every asset's depreciation roll-forward, the rental sale's §1250 recapture, CTC phase-out, and passive-loss release on the sale.
